

Iteration in Programming

while loops and loop control variable

Produced
by:

Dr. Siobhán Drohan,
Ms. Mairead Meagher,
Ms. Siobhán Roche.

Topics list

1. Repetition in Programming – Intro to looping
2. Use of loops (while loops).

Recap: Boolean conditions

- A boolean condition is an expression that evaluates to either true or false e.g.

$x < 50$

- Boolean conditions can be used to control:
 - Selection i.e. if statements and
 - Iteration i.e. loops (we will look at these now).

Repetition in Programming

- Computers are very good at repetition.
- Example:
 - `calculate pay` for 1000 employees.
 - You should use the same `calculate pay` algorithm 1000 times.
 - You don't write the `calculate pay` algorithm 1000 times; instead you include it in a loop.

Form of loop

- Write the all the numbers between 1 and 5 inclusive
 - Without loop:

```
System.out.println(1);  
System.out.println(2);  
System.out.println(3);  
System.out.println(4);  
System.out.println(5);
```

Form of loop

- Write the all the numbers between 1 and 5 inclusive
 - With a loop:
 - do this 5 times

`System.out.println(i);`

With the value of i going from 1 to 5

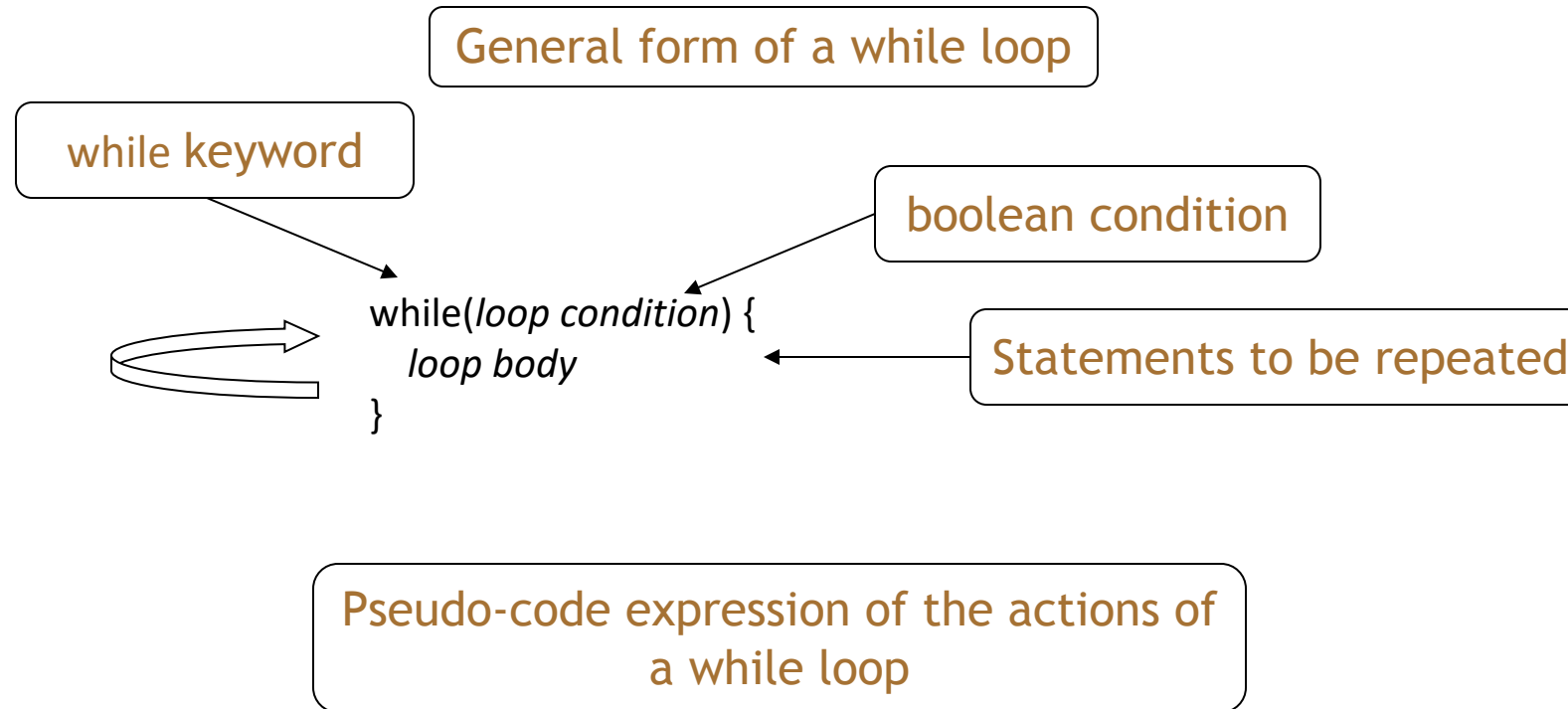
Topics list

1. Repetition in Programming – Intro to looping
2. Use of loops (while loops).

Loops in Programming

- There are three types of loop in (Java) programming:
 - **while** loops
 - **for** loops
 - **do while** loops

while loop pseudo code



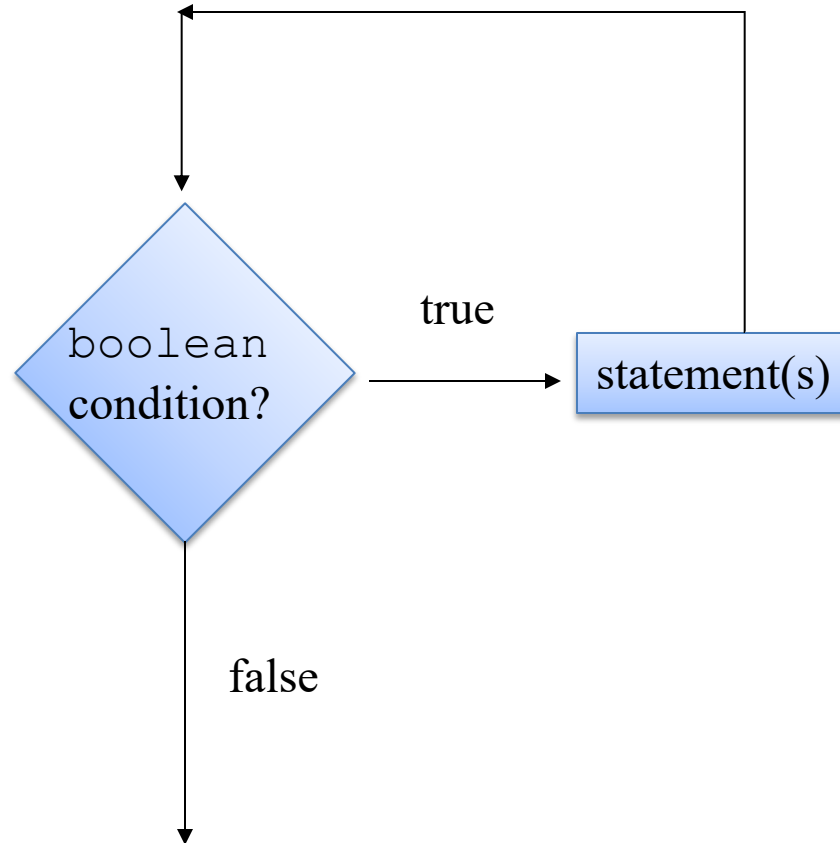
while we wish to continue, do the things in the loop body

Construction of while loop

```
Declare and initialise loop control variable (LCV)
while(boolean condition based on LCV is true)
{
    "do the job to be repeated"
    "update the LCV"
}
```

This structure should always be used

while loop Flowchart



```
int i = 1;      //i is the LCV
while(i <=5)
{
    System.out.println(i);
    i++;
}
```

Processing Example 4.5

```
int i = 1; //i is the LCV  
while(i <=5)  
{  
    System.out.println(i);  
    i++;  
}
```

Declare and initialise **loop control variable (LCV)**

while(boolean condition based on **LCV** is true)

{

“do the job to be repeated”

“update the **LCV**”

}



```
1  
2  
3  
4  
5
```

Processing Example 4.5

1. Declare and initialise **loop control variable (LCV)**

```
int i = 1;  //i is the LCV
while(i <=5)
{
    System.out.println(i);
    i++;
}
```

2. boolean condition based on **LCV**

“do the job to be repeated”

“update the **LCV**”

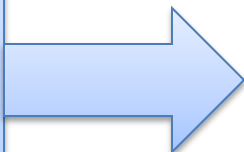
1
2
3
4
5

Some Study Exercises

This basic while loop,
produces this output.

```
int i = 1;

while (i <=5)
{
    System.out.println("Hello World");
    i++;
}
```



```
Hello World
Hello World
Hello World
Hello World
Hello World
```

Some Study Exercises

1. Change the code so that “Hello World” is printed out 10 times.
2. Change the code so that the numbers from 1 to 10 (inclusive) are printed out, one line at a time.
3. Change the code so that the numbers from 10 to 1 are printed out.

Questions?

